

Kevin Pierce Ramos

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Software engineer with hands-on production experience across embedded systems, application development, and cloud infrastructure. Professional background includes C++/Python development for robotic manipulation systems, hardware integration and diagnostics on Linux platforms, and building an operator-facing Android control application in Kotlin. Project experience spans cloud-native REST APIs with CI/CD pipelines, network security appliances, and systems-level tooling. Holds a B.S. in Computer Science from UNLV and begins an M.S. in Computer Science at Georgia Tech in Spring 2027.

PROFESSIONAL EXPERIENCE

Richtech Robotics, Las Vegas, NV

Oct 2025 – Present

Technical Solution Specialist

- Engineered and supported dual 6-DOF robotic manipulation systems in live production environments, developing C++ and Python automation logic for motion control, calibration routines, and hardware diagnostics across Linux-based platforms.
- Contributed to a Kotlin Multiplatform and Jetpack Compose Android tablet application that serves as the operator control interface for a Richtech robot, building UI screens, networking, and business logic using Retrofit/OkHttp, a repository pattern, Koin dependency injection, and MVVM ViewModels.
- Integrated and troubleshoot industrial serial communication interfaces (RS-232/RS-485) connecting servo drives, motor controllers, and distributed hardware components; performed system bring-up, signal validation, and fault diagnosis.
- Built observability and monitoring tooling from scratch, including structured logging, real-time metrics collection, and fault alerting, to surface hardware and software performance issues proactively; collaborated with mechanical and electrical engineers on integration and delivery.

PROJECTS

Event Analytics API · *FastAPI, PostgreSQL, Docker, Terraform, GitHub Actions, AWS EC2/S3, SQLAlchemy, Alembic, pytest*

- Designed and built a production-ready REST API featuring time-series analytics, filtered queries, pagination, and schema migration via Alembic. Containerized with Docker, provisioned AWS infrastructure with Terraform, and implemented a full GitHub Actions CI/CD pipeline with automated testing, image build, and deployment, plus structured logging for CloudWatch-compatible observability.

Travel Pi: Network Security Appliance · *Linux, Bash, WireGuard, iptables/UFW, Pi-hole, Tailscale, systemd, hostapd*

- Provisioned and configured a portable Linux network security appliance end to end: WireGuard VPN tunneling, stateful firewall rules via iptables/UFW, DNS-layer filtering with Pi-hole, wireless hotspot bridging via hostapd, and Tailscale mesh networking. Wrote idempotent Bash provisioning scripts and systemd health-check services for self-healing headless operation.

EulogyQuest: Full-Stack Web Application · *React, Node.js/Express, PostgreSQL, Docker, JWT Auth*

- Built a full-stack capstone application with a React frontend and Node.js/Express backend, implementing JWT-based authentication, RESTful API design, and PostgreSQL persistence. Containerized the full stack with Docker Compose for consistent local and production environments.

Logstat · *C++17, CMake, Linux*

- Built a high-performance C++ CLI tool for streaming log analysis, computing per-service latency percentiles and error-rate distributions from large datasets in real time and outputting structured JSON diagnostic reports. Demonstrates systems-level C++ proficiency and performance-oriented design on Linux.

Espresso Compiler · *Java, JFlex, Java CUP*

- Implemented a full compiler pipeline from scratch: hand-written lexer, LALR parser, AST construction with visitor pattern, symbol table management, type checking, and modifier validation with structured error reporting and a comprehensive test harness.

TECHNICAL SKILLS

Languages: Python, C++, Kotlin, Java, JavaScript, Bash, Go, SQL, HTML/CSS

Embedded & Hardware: Embedded Linux, real-time control loops, hardware bring-up, motor controllers and servo drives, sensors, RS-232/RS-485 serial interfaces, system integration
Mobile & UI: Kotlin Multiplatform, Jetpack Compose, MVVM, Retrofit/OkHttp, Koin, kotlinx.serialization, DataStore, React
Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, Git, CI/CD
Backend & APIs: FastAPI, REST, Node.js/Express, PostgreSQL, SQLAlchemy, Docker Compose
Networking & Security: TCP/IP, DNS, SSH, WireGuard/IPsec, iptables/UFW, VPN, Linux networking
Systems: Linux (Ubuntu/Debian/embedded), systemd, Bash scripting, CMake, GDB, process management

EDUCATION

Georgia Institute of Technology , Atlanta, GA M.S. in Computer Science	<i>Starting Spring 2027</i>
University of Nevada, Las Vegas , Las Vegas, NV B.S. in Computer Science	<i>Graduated May 2025</i>